Conditional LoRA Bridges at 230M: What Replicates, What Softens, and What Inverts

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2026-07-03

Abstract

Our controlled study of conditional LoRA bridges [1] established its findings on a purpose-trained 33.6M-parameter model, leaving open which conclusions are properties of the mechanism and which are artifacts of scale. We port the unchanged experiment suite to a pretrained 230M-parameter model (LiquidAI LFM 2.5) and rerun it at both the standard and the 10× training budget. The methodological core replicates exactly: bits-per-byte cannot detect sensor conditioning (the capacity-matched constant-feature control matches the true-feature bridge, +0.0852 vs +0.0856), while task-aligned probes detect it decisively (0.72 rank-1 on six-way IMU, 0.31 on fifty-way audio, all controls at chance). Three scale-dependent results reverse or soften. Prefix tuning flips sign, from -0.0048 BPB on the small model to +0.0614 at 230M, while remaining the weakest adapter. The diversity–BPB tradeoff softens: the strongest penalty retains 89% of the unregularized BPB gain where the small model retained none. Most strikingly, the long-budget collapse inverts: extended unregularized task training destroyed IMU conditioning on the small model (rank-1 0.80 to 0.17, chance) but strengthens it at 230M (0.72 to 0.89), even as weight collapse under pure text loss persists at both scales. At the two scales tested, capacity and pretraining are allies of task-conditioned weight-space bridges, and single-scale claims about the mechanism's failure modes should be read as lower bounds.

1 Motivation

The conditional-LoRA-bridges study [1] reported one sharp methodological result (aggregate language-modeling metrics cannot certify sensor conditioning) and one mechanism result, hypernetwork weight collapse, with facets that include degradation from token-space (prefix) conditioning and a budget-dependent destruction of task-level conditioning under prolonged unregularized training. All of it was measured on a single 33.6M-parameter model trained from scratch. This note answers the obvious question: which of those findings survive contact with a real pretrained model?

2 Porting the harness

We add a checkpoint adapter (`hf:<model-id>`) to the published harness so that every experiment (baselines, controls, probes, budgets, seeds) runs unchanged against a HuggingFace causal LM; here LiquidAI LFM 2.5 230M [2], a 14-layer hybrid short-convolution/attention model with a 65,536-token vocabulary. Three porting details matter for fidelity:

Table 1: BPB improvement over the frozen baseline (mean across audio, vision, IMU; standard budget; seed-42 protocol as in the original paper). The decomposition — bridge ≈ constant control < static LoRA — replicates; prefix tuning flips sign.

Method	33.6M (paper)	230M (this note)
Static LoRA	+0.0033	+0.0876
Conditional bridge	+0.0027	+0.0856
Shuffled features	+0.0018	+0.0853
Random features	+0.0006	+0.0823
Constant features	+0.0027	+0.0852
Prefix tuning	-0.0048	+0.0614

- **LoRA targets.** The paper's target set "all" (every attention and MLP linear) maps to q/k/v/out projections on the six attention layers and the SwiGLU projections on all fourteen layers; short-convolution projections are left unadapted. The generated LoRA vector has $D = 774{,}144$ dimensions at rank 4.

- **Byte-exact BPB across tokenizers.** Bits-per-byte is tokenizer-independent only if the per-token byte table reconstructs the exact UTF-8 length of any encoded text. We build and validate such a table for the model's byte-level BPE, including its non-special added tokens, so BPB values are comparable across the two vocabularies.

- **Sequence-start sensitivity.** LFM 2.5 is strongly BOS-dependent (7.13 nats/token without BOS vs. 2.70 with, on the same text). The packing dataloader already places BOS at every row start; the task probes prepend BOS for `hf:` models so the probe measures conditioning, not an out-of-distribution artifact. The published small-model protocol is unchanged and reproduces bit-identically.

The frozen 230M model evaluates at 1.1839 BPB on the paper's validation stream (the small model: 1.9654).

3 What replicates

Table 1 gives the BPB decomposition at the standard budget. The paper's central methodological claim transfers unchanged: the conditional bridge (+0.0856) is indistinguishable from the capacity-matched constant-feature control (+0.0852), static LoRA is the best pure-text adapter (+0.0876), and three-seed spreads are below 10^{-3}. BPB still cannot see conditioning; the task probes still can (Table 2): true features reach 0.72 rank-1 on six-way IMU and 0.31 on fifty-way audio while shuffled, random, and no-bridge controls sit at chance, numerically close to the small model's 0.80 and 0.33. Composition of independently trained bridges remains sub-additive at both scales, and unregularized text-loss training still collapses generated weights toward input-independence (cross-input cosine 0.99 at $\lambda = 0$).

4 What softens

Prefix tuning flips sign. On the small model, feature-conditioned prefix tuning strictly degraded BPB (-0.0048); at 230M it improves it (+0.0614), yet remains the weakest adapter by a

Table 2: Task-probe rank-1 accuracy for true features vs. the no-bridge reference, at both budgets. The small model's IMU probe collapses to chance at the 10× budget; the 230M model's strengthens.

			33.6M		230M	
Probe	Budget	Chance	true	none	true	none
IMU	standard	0.17	0.80	0.17	0.72	0.16
IMU	10×	0.17	0.17	0.17	0.89	0.16
AUDIO	standard	0.02	0.33	0.03	0.31	0.02
AUDIO	10×	0.02	0.36	0.03	0.61	0.02

Table 3: Cross-input cosine of generated weights (IMU; lower = more input-dependent) across the diversity sweep, at both scales and budgets.

λ	33.6M std	33.6M 10×	230M std	230M 10×
0.00	0.97	0.99	0.99	0.98
0.01	0.92	0.99	0.98	0.98
0.05	0.65	0.31	0.96	0.98
0.10	0.44	0.47	0.60	0.68
0.20	0.55	0.17	0.72	0.36

wide margin against the $\sim$+0.0856 weight-space rows. The original claim is therefore correctly scoped to small models trained without prefixes, while the paper's ordering, in which weight-space conditioning dominates token-space conditioning, holds at both scales.

The diversity–BPB tradeoff softens. On the small model, enforcing input-dependent weights at $\lambda = 0.20$ erased the BPB gain entirely; at 230M the same penalty retains 89% of it (+0.0767 of +0.0860). The non-monotonicity the paper reports at the largest tested weight (cross-input cosine reversing at $\lambda = 0.20$) appears at the standard budget at 230M as well (0.60 at $\lambda = 0.10$ vs. 0.72 at $\lambda = 0.20$), but at the 10× budget the reversal disappears and cross-input cosine falls to 0.36 at the largest weight (Table 3). The penalty required for input-dependence grows with both budget and scale: on the small model $\lambda = 0.01$ loses its effect at the long budget, while at 230M penalties below $\lambda = 0.10$ never produce input-dependent weights at either budget.

5 What inverts

The sharpest warning in the original paper was budget-dependent: at 10× training, the small model's unregularized IMU bridge became behaviorally indistinguishable from no bridge at all (rank-1 0.17, equal to chance), having scored 0.80 at the standard budget. At 230M the same protocol *strengthens* conditioning: IMU rank-1 rises from 0.72 to 0.89 (top-2 1.00), and audio rises from 0.31 to 0.61 against a 0.02 chance level, with every control still at chance. Prolonged task-aligned training is an ally of conditioning at this scale, not a failure mode. The weight-space collapse under pure text loss still occurs at 230M (Table 3, $\lambda = 0$); what changes is that label-paired training at scale maintains input dependence without regularization over the budgets tested.

6 Implications

Three practical updates for anyone building on the paper. First, the evaluation discipline (capacity-matched controls, task-aligned probes) transfers verbatim and remains necessary: BPB is exactly as blind at 230M as at 33.6M. Second, the mechanism's failure modes shift in the favorable direction from the 33.6M model to the 230M one: the task-conditioning collapse threshold moves outward and the cost of enforcing input-dependent weights shrinks. Two caveats bound that statement: the comparison confounds scale with pretraining (the small model was trained from scratch, LFM 2.5 is pretrained), and it concerns task-conditioned training only, since collapse under pure text loss is unchanged at 230M. Even so, claims about weight-space conditioning's fragility measured at one scale should be read as lower bounds. Third, token-space and weight-space conditioning both improve with scale, but the gap between them persists; weight-space modulation remains the stronger interface across every configuration tested.

Reproducibility

The port is additive: the published harness reproduces the original paper bit-identically, and the same runner produces every number here via `--checkpoint hf:LiquidAI/LFM2.5-230M`. Artifacts: the standard-budget suite (35 experiments) and the 10× suite (33 of 35; the two remaining seed-stability repetitions were not run) ship alongside this note as `summary-lfm.json` and `summary-lfm-scaling.json` in the code repository, with the original paper's artifacts unchanged. Code: https://github.com/kortexa-ai/legolm.paper.

References

[1] F. Penov. Conditional LoRA Bridges for Modular Sensor Adaptation of Frozen Small Language Models. Preprint, 2026. https://github.com/kortexa-ai/legolm.paper.

[2] Liquid AI. LFM2.5-230M. Model release, 2025–2026. https://huggingface.co/LiquidAI/LFM2.5-230M.